

ACADEMIC DETAILS

Degree	Specialization	Institute	Year	CPI/%
B.Tech.	Mechanical Engineering	IIT Gandhinagar	2024-2028	8.30
Class XII	Physics, Chemistry, Maths	Cosmos Castle International School	2023-2024	88.8%
Class X		Navkar Public School	2021-2022	93.5%

INTERNSHIPS

- **Robotics Engineering Intern | Krishna Defence & Allied Industries Limited**

[May 2026-July 2026]

(Research & Development | Gandhinagar, Gujarat) | [Project Link](#)

- Engineered an autonomous smart forklift localization pipeline in ROS 2, developing custom sensor fusion algorithms to integrate UWB ranging and IMU telemetry
 - Optimized real-time indoor navigation by mitigating UWB signal noise and sensor drift, achieving highly precise state estimation and positional tracking

PROJECTS

- **Autonomous UR5e Control via Deep RL (MuJoCo)**

[December 2025 & June 2026-July 2026]

[Project Link](#)

- Engineered a custom Gymnasium environment bridging MuJoCo physics to train a 6-DOF UR5e industrial manipulator from scratch using Proximal Policy Optimization (PPO)
 - Implemented rigid body dynamics, inverse kinematics using Mocap bodies, and continuous action-space control using Python, NumPy, and Stable-Baselines3
- **Design and Kinematic Analysis of a Crank-Rocker Mechanism**

[October 2025 - November 2025]

(Advisor:- Prof. Jayaprakash) | [Project Link](#)

- Designed and fabricated a low-friction planar four-bar crank-rocker mechanism, and developed a custom MATLAB script to automate theoretical position and velocity analysis using Grashof's Law and vector loop equations
 - Validated the theoretical model through experimental video tracking, engineering a Python script to process synchronized position data, apply numerical differentiation, and perform comprehensive error analysis
- **Validation of Angular Momentum Conservation via Reynolds Transport Theorem**

[January 2026 - April 2026]

(Advisor:- Prof. Shubhangi Bansude) | [Project Link](#)

- Designed and fabricated a three-arm rotating water sprinkler apparatus using commercially available components, and developed the complete plumbing and support system for controlled experimental testing
 - Experimentally validated the Reynolds Transport Theorem and conservation of angular momentum by comparing theoretical predictions with experimental observations under multiple operating conditions, and performed uncertainty analysis using the Root Sum Square (RSS) method

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++

• **Libraries:** Pandas, Numpy, Matplotlib

• **Tools:** Autodesk, Ansys (Basic), GitHub, MSC Adams (Basic), MATLAB, Mathematica

• **Robotics:** ROS 2, MuJoCo

POSITIONS OF RESPONSIBILITY

- **Coordinator, Tinkerer's Lab**

[May 2026 - Present]

- Organized and coordinated technical workshops and innovation-driven events, ensuring seamless planning, execution, and participant engagement
 - Conducting events of Tinkerer's Lab and improving or resolving the problems and features of the Lab
- **Sponsorship Team Senior Member, Blithchron IITGN**

[August 2025 - February 2026]

- Conducted mock interview sessions for juniors, offering guidance on company recruitment processes

ACHIEVEMENTS AND EXTRACURRICULAR ACTIVITIES

- Secured All India Rank (AIR) 3630 in JEE Mains and AIR 5433 in JEE Advanced

• Secured 1st place among 50+ teams in the Hallabol Chess Tournament at IIT Gandhinagar as a member of the winning team

• Secured the Gold Medal and Championship Trophy at the KnightPlayers Grand Chess Championship, IITGN; played in all tournament fixtures and was awarded Man of the Match in the finals